FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Rational Use of Investigations

Lecture | Week 4 · Small babies, sugar & the heart | Foundations | Dr Nabatte

A test is worth doing only if its result will change what you do for the child. Rational investigation means estimating the pre-test probability, choosing the cheapest test that answers the question, interpreting results against age-specific normal ranges, and never delaying emergency treatment for a result. Malaria RDTs, FBC, glucose, CSF, urine, HIV PCR and TB tests in children are examined every year.

**Must know before the ward round**

1. **Ask before every test: will the result change my management?** If not, do not order it. Never delay emergency treatment (glucose, artesunate, antibiotics, oxygen) while waiting for a result.
2. **Sensitivity rules out (SnOut), specificity rules in (SpIn). PPV and NPV depend on prevalence**: the same mRDT result means more in a high-transmission district than in a low one.
3. **Bedside tests in every sick child (ETAT)**: **glucose**, **Hb (HemoCue)**, **malaria RDT**, **$SpO_2$**. Hypoglycaemia: <2.5 mmol/L (WHO) / <3.0 (UCG, SAM); neonates <2.6 mmol/L.
4. **HRP2 malaria RDTs stay positive for 2–5 weeks after successful treatment**: use **microscopy** to judge treatment response and recurrent fever after recent treatment.
5. **HIV in children <18 months needs a DNA PCR** (maternal antibody makes antibody tests positive). Uganda EID: PCR at **6 weeks**, **9 months**, and **6 weeks after breastfeeding stops**; rapid test at **18 months**.
6. **TB in children**: Xpert MTB/RIF Ultra on **sputum, gastric aspirate, nasopharyngeal aspirate or stool**; **urine LF-LAM** in children with HIV; a negative test does not exclude TB (use the WHO 2022 treatment-decision algorithm).
7. **Normal CSF after 1 month**: **≤5 lymphocytes, 0 neutrophils** × $10^6$/L, protein **<0.4 g/L**, CSF:blood glucose **≥0.6**. Neonates: up to ~20 cells and protein <1.0 g/L. **Bacterial**: neutrophils 100–10,000, protein >1, ratio <0.4.
8. **Age matters for normal ranges**: Hb nadir 9–11 g/dL at 2 months (physiological); **lymphocytes predominate** from about 1 week to 5 years; **ALP is high in growing children**; creatinine is low (≈20–35 µmol/L in infants).
9. **Urine for UTI**: clean-catch or catheter, never a bag for culture; **under 3 months do microscopy and culture**, not dipstick alone (NICE NG224, 2022). Nitrite is specific but insensitive in infants.
10. **Blood volume matters**: a neonate has only **80–90 mL/kg** of blood; 3 mL of samples from a 1 kg baby is about 3.5% of its blood volume. Take the smallest volumes, batch tests, and avoid daily routine bloods.
11. **Febrile convulsion, bronchiolitis, uncomplicated pneumonia and gastroenteritis need few tests**: glucose and malaria test; no routine CXR, CT or electrolytes unless the child is severely ill or atypical.
12. **Sickling test** misses disease under 6 months (high HbF) and cannot separate trait from disease: confirm with **Hb electrophoresis/HPLC** or a validated point-of-care test.


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# In a nutshell

Every investigation is a question you ask the child's body. A good question has an answer that **changes what you do**. Rational use of investigations means choosing **the right test, for the right child, at the right time, at the right level of care**, and interpreting it properly. In Uganda this matters more than anywhere: families often pay for tests out of pocket, laboratories at many facilities are thin, results come back slowly, and blood is precious in a small, anaemic child.

Over-investigation wastes money, delays treatment, takes blood from children who have little to spare, exposes them to radiation and produces false positives that lead to unnecessary treatment. Under-investigation misses treatable disease: the mRDT-negative child with meningitis, the "malaria" that was a UTI, the HIV-exposed infant never tested.

**CLINICAL PEARL**

Before you write any request form, say three things to yourself: **"What do I think this is (and how likely)? What will this test show if I am right? What will I do differently with that result?"** If you cannot answer the third question, do not send the test.

**UGANDA CONTEXT**

This topic follows the **Uganda Clinical Guidelines (UCG) 2023**, the **WHO Pocket Book of Hospital Care for Children (2013)**, the **WHO 2024 pneumonia and diarrhoea guideline**, the **WHO guidelines for malaria (September 2026 edition)**, the **WHO 2022 TB guidelines for children and adolescents**, the **Uganda consolidated HIV guidelines (2022)** and **NICE** guidance on fever (NG143) and UTI (NG224). Laboratory capacity follows the Uganda tiered system (HC II to national referral) and the **national sample-transport (hub) network** that carries samples such as EID dried blood spots and Xpert specimens to testing sites.

**EXAM FAVOURITE**

The classic viva: "A febrile 2-year-old has a negative malaria RDT. What do you do?" **Believe a good-quality negative RDT** (sensitivity >95% for falciparum at clinical parasite densities), **do not give antimalarials**, and **look for another cause** (pneumonia, otitis, UTI, meningitis, typhoid, measles). Repeat the test or do microscopy only if the illness is severe or the clinical picture strongly suggests malaria.

# Definitions & classification

| Term | Definition | Bedside meaning |
|---|---|---|
| **Sensitivity** | Proportion of people **with** the disease who test positive | A very sensitive test, when **negative**, helps **rule out** (SnOut) |
| **Specificity** | Proportion of people **without** the disease who test negative | A very specific test, when **positive**, helps **rule in** (SpIn) |

| Term | Definition | Bedside meaning |
|---|---|---|
| **Positive predictive value (PPV)** | Proportion of positive tests that are true positives | **Rises with prevalence**: falls when you test low-risk children |
| **Negative predictive value (NPV)** | Proportion of negative tests that are true negatives | **Falls as prevalence rises** |
| **Pre-test probability** | How likely the disease is before the test (from prevalence, history and examination) | The single most important factor in interpreting a result |
| **Likelihood ratio (LR)** | LR+ = sensitivity ÷ (1 – specificity); LR– = (1 – sensitivity) ÷ specificity | LR+ >10 or LR– <0.1 shift probability a lot; LR 0.5–2 barely moves it |
| **Post-test probability** | Probability after the result | Pre-test odds × LR = post-test odds |
| **Reference interval ("normal range")** | The central 95% of results in healthy people of that age | **5% of healthy children are "abnormal"** on any one test; with 20 tests, most healthy children have at least one abnormal value |
| **Screening test** | Testing apparently well people to find early disease (e.g. newborn sickle cell screening, HIV testing of exposed infants) | Needs high sensitivity, low cost, and an effective intervention |
| **Diagnostic test** | Testing a child with symptoms to confirm or exclude a diagnosis | Chosen by pre-test probability |
| **Monitoring test** | Repeated to track disease or treatment (viral load, Hb in severe malaria, sodium in DKA) | Only as often as the result can change management |
| **Point-of-care test (POCT)** | A test done at the bedside with an immediate result (glucose, HemoCue Hb, mRDT, HIV RDT, urine dipstick, POC CD4, TB LAM) | Ideal for low-resource settings when quality-assured |
| **Gold (reference) standard** | The best available test for a condition (blood culture for bacteraemia, CSF culture for meningitis, Hb electrophoresis for SCD) | Often slow, expensive or unavailable |

### Types of investigation by purpose

1. **Diagnostic**: does the child have malaria, UTI, TB, HIV?
2. **Severity and complications**: how sick is this child? (glucose, Hb, $SpO_2$, electrolytes, creatinine, lactate)
3. **Baseline before treatment**: creatinine before gentamicin in a sick child, LFTs before TB drugs in liver disease, Hb before AZT.
4. **Monitoring response and toxicity**: Hb after transfusion, sodium in DKA, viral load on ART.
5. **Screening**: newborn sickle cell screening, HIV testing of exposed infants, TB contact screening, MUAC screening.

## Epidemiology & Uganda context

- **What is available where (typical)**:

| Level | Tests usually available |
|---|---|
| **VHT / HC II** | Malaria RDT, MUAC tape, ORS-based assessment, HIV rapid test (outreach), respiratory rate timer |

| Level | Tests usually available |
|---|---|
| **HC III** | Malaria microscopy and RDT, HIV RDT, Hb (HemoCue or colorimetric), urine dipstick, pregnancy test, RPR/syphilis RDT, sputum smear or sample referral for Xpert, glucose (often) |
| **HC IV / general hospital** | FBC (sometimes), blood grouping and cross-match, Xpert MTB/RIF, CD4/POC CD4, basic biochemistry (variable), ultrasound and X-ray (variable), CSF microscopy and Gram stain, sickling test |
| **Regional referral hospital** | FBC, U&E, creatinine, LFTs, CRP (variable), blood culture (some), X-ray, ultrasound, ECG, sometimes echocardiography and CT |
| **National referral (Mulago, Mengo, Lubaga, Nsambya, Mbarara/Gulu teaching hospitals)** | Cultures and sensitivities, CT, MRI (Mulago), Hb electrophoresis/HPLC, blood gases, echocardiography, TB culture and line-probe assays (via national TB reference lab), HIV viral load and EID PCR (via CPHL), histology |

- **The hub system**: samples for **HIV DNA PCR (dried blood spots), viral load, Xpert and TB culture** travel from lower facilities via **district hubs** to the **Central Public Health Laboratories (CPHL)** and the **National TB Reference Laboratory**. Results return by SMS/printer or the hub rider. **Turnaround of days to weeks** means you must **track results** and recall the family; many EID results are never collected.
- **Out-of-pocket costs**: in private wings and private labs, families pay for every test. A "full panel" on a febrile child can cost more than the treatment. Public facilities are free in principle but often out of reagents, sending families to private labs.
- **Overuse is common**: routine "widal" tests for typhoid (poor sensitivity and specificity in endemic areas), stool microscopy "for worms" in every child, repeated malaria smears after an RDT-confirmed diagnosis, and routine CXR for every cough.
- **Underuse is also common**: blood glucose not checked in convulsing children, HIV status unknown on discharge, urine never tested in febrile infants, and TB never investigated in a child with weight loss.

# Principles

### 1. Probability thinking (Bayes at the bedside)

A result changes your estimate; it rarely gives certainty. Start with the **pre-test probability** (from local prevalence + history + examination), and let the test move it.

**Worked example: malaria RDT in two settings** (assume sensitivity 95%, specificity 95%)

| Setting | Pre-test probability | Of 1,000 febrile children: true + / false + | PPV | NPV |
|---|---|---|---|---|
| High-transmission district (e.g. northern Uganda, rainy season) | 50% | 475 / 25 | **95%** | 95% |
| Low-transmission Kampala suburb | 5% | 47.5 / 47.5 | **50%** | >99.7% |

In Kampala a positive RDT is a coin toss for being the true cause, while a **negative RDT almost excludes malaria**. In a high-transmission area, a positive RDT is usually right, but **asymptomatic parasitaemia is common**: the child may have malaria parasites *and* pneumonia or meningitis. Always examine for other causes.

### 2. Test thresholds

- **Treatment threshold**: the probability above which you would treat without testing (e.g. a convulsing, comatose febrile child in a high-transmission area gets **IV artesunate immediately**, test or no test).

- **Testing threshold**: the probability below which you would neither test nor treat (e.g. a well child with a cold in Kampala does not need a malaria test unless febrile).
- **Test when the probability lies between the two** and the test is good enough to move it across a threshold.

## 3. Every test has harms

| Harm | Example |
|---|---|
| **Blood loss** | Daily FBC, U&E and CRP in a preterm baby → iatrogenic anaemia and transfusion |
| **Pain and distress** | Repeated venepuncture, LP; children remember |
| **Delay** | Waiting for a CT before LP or antibiotics in suspected meningitis; waiting for a smear before treating severe malaria |
| **False positives** | Widal "positive" → unnecessary antibiotics; HRP2 RDT positive after recent treatment → unnecessary antimalarials; contaminated blood culture (coagulase-negative staph) → vancomycin |
| **Overdiagnosis / incidentalomas** | A CT done for minor head injury finds an arachnoid cyst → anxiety and follow-up scans |
| **Radiation** | A head CT ≈ 1–2 mSv (about 6–12 months of background radiation); children are more radiosensitive; repeated CTs measurably increase cancer risk |
| **Cost** | Money spent on tests is money not spent on drugs, food or transport back for review |
| **Stigma / disclosure** | HIV or TB testing without proper counselling and consent |

## 4. Blood volume in small children

- Total blood volume: **neonate 80–90 mL/kg**, child about **80 mL/kg**, adult 70 mL/kg (UCG table). A 1 kg preterm has about **85 mL** of blood; a 1 mL sample is >1% of it.
- Common limits: take **no more than about 1–5% of blood volume in 24 hours** and **about 10% over 8 weeks** for non-urgent tests.
- **Batch tests**, use **paediatric (microtainer) tubes**, use **capillary samples** for glucose, Hb, bilirubin and some chemistry, and record the volume taken in neonates.

## 5. Sample quality makes or breaks the result

| Problem | Effect | Prevention |
|---|---|---|
| **Haemolysis** (squeezed heel, small needle, forceful syringe) | **Falsely high potassium**, high LDH, interferes with bilirubin | Free-flowing samples, wider needle, gentle transfer |
| **Sample taken from the drip arm** | High glucose, low Hb and electrolytes diluted | Sample from the other arm or stop the drip |
| **Delay before glucose measurement** | Glycolysis lowers glucose **≈5–7% per hour** at room temperature | Measure at the bedside or use fluoride tubes |
| **Clotted EDTA sample** | Falsely low platelets | Mix gently, fill to the line |
| **EDTA contamination of biochemistry tube** | **High K, very low Ca** | Correct order of draw |
| **Bag urine** | Contamination (false-positive UTI) | Clean-catch, catheter or suprapubic aspirate for culture |
| **Blood culture underfilled or contaminated** | Missed bacteraemia or false positives | 1–3 mL in infants (≥1 mL), skin antisepsis, single venepuncture |
| **Unlabelled or mislabelled sample** | Wrong patient's result → wrong treatment | Label at the bedside, two identifiers |

### 6. Normal values change with age

| Test | Neonate | Infant | Child | Comment |
|---|---|---|---|---|
| **Hb (g/dL)** | 14–22 at birth | **Nadir 9–11 at 2–3 months** (physiological anaemia) | Anaemia (WHO 2024): **<10.5 at 6–23 months; <11.0 at 24–59 months; <11.5 at 5–11 years**; <12.0 at 12–14 years | Do not transfuse a well 2-month-old with Hb 9.5 |
| **WBC (× $10^9$/L)** | 9–30 (first days) | 6–18 | 5–15 | **Lymphocyte predominance** from about **1 week to 4–5 years**; neutrophil predominance before and after |
| **Platelets (× $10^9$/L)** | 150–450 | 150–450 | 150–450 | Low in malaria, sepsis, HIV, DIC |
| **Creatinine (µmol/L)** | Reflects mother's in first days | ≈20–35 | ≈25–60 (rising with muscle mass) | "Normal adult" creatinine of 80 in a toddler is **renal failure** |
| **ALP (U/L)** | High | High | **High in growth spurts** (up to 2–4× adult) | Not liver disease; very high or with low phosphate → rickets |
| **Glucose (mmol/L)** | Treat **<2.6** | Treat <2.5 (WHO), <3.0 (UCG) | Same | SAM: treat <3.0 |
| **Heart rate, RR, BP** | See General Examination topic | | | Age-specific norms |

# Clinical acumen

## Focused history

The history sets the pre-test probability, which decides what to test. Ask:

- **Duration of fever and pattern**: fever ≥7 days → typhoid, TB, UTI, abscess, HIV, endocarditis, malignancy; brief fever with coryza → viral.
- **Previous antimalarial or antibiotic treatment** (and when): HRP2 RDTs remain positive for weeks; partial antibiotic treatment changes CSF and cultures.
- **Danger signs**: convulsions, unable to drink, vomiting everything, lethargy → bedside glucose, Hb, malaria test, consider LP and blood culture.
- **Exposure**: TB contact (household cough), HIV status of mother and breastfeeding, sick contacts with diarrhoea, travel to a high-malaria area, animals (brucellosis), unsafe water (typhoid, cholera).
- **Urinary symptoms** in older children; foul urine, poor feeding or fever without a focus in infants → urine.
- **Bleeding, bruising, pallor, bone pain**: FBC and film (leukaemia, sickle cell, ITP).
- **Weight loss and night sweats**: TB, HIV, malignancy.
- **Immunisation status**: an unimmunised child with fever and cough may have measles or pertussis; Hib/PCV coverage changes meningitis risk.
- **What tests have already been done**, at which facility, and the results (avoid repeating a valid test).

## Examination

- **ABCD and vital signs**: decide urgency (bedside tests now, the rest later).
- **Pallor** (palms, conjunctivae): Hb now; severe pallor plus distress → cross-match.
- **Jaundice**: in neonates, bilirubin (TcB/TSB) with the hour of life; in older children, think malaria haemolysis, sickle cell, hepatitis.
- **Focus of infection**: ears, throat, chest, abdomen, joints, skin, meninges (neck stiffness, bulging fontanelle), urine. A clear focus often means fewer tests.
- **Signs of chronic illness**: wasting, stunting, clubbing, lymphadenopathy, hepatosplenomegaly, oral thrush (HIV), BCG scar.
- **Neurology**: focal signs, papilloedema or deteriorating consciousness change the LP decision.

## Red flags

RED FLAG

- **Do not delay treatment for tests** in: shock, hypoglycaemia, severe malaria, suspected meningitis or sepsis, status epilepticus, severe pneumonia with hypoxaemia. Take a quick sample if you can (glucose, Hb, RDT, blood culture), then treat.
- **A critical result** (glucose <2.5, K >6.5 or <2.5, Na <125 or >155, Hb <5, platelets <20, positive blood culture or CSF Gram stain) must be **acted on immediately** and communicated to the clinician by phone, not left in a pile.
- **A result that does not fit the child** ($K^+$ 7.8 in a well, chatty child; Hb 3 in a pink, active child): **repeat before acting** (haemolysis, drip-arm sample, wrong patient).
- **LP is contraindicated** (or delayed) with: signs of raised intracranial pressure (papilloedema, focal signs, abnormal posturing, irregular breathing, unequal pupils), coma with GCS <8 or deteriorating consciousness, shock, coagulopathy/low platelets, skin infection over the site. **Give antibiotics first.**
- **A negative test that contradicts a strong clinical picture** (negative Xpert in a child with classic TB; negative RDT in a comatose child with retinopathy): treat on clinical grounds and seek more evidence.

## Thinking like a clinician

- **"Will this result change what I do?"** A FBC in a child with uncomplicated malaria rarely changes management; an Hb does if pale.
- **"What will I do if it is positive, and if it is negative?"** If you would do the same either way, don't send it.
- **"Is there a cheaper, faster bedside test that answers the question?"** HemoCue instead of full FBC for Hb; urine dipstick in older children; RDT instead of microscopy for diagnosis.
- **"Is this a screening question or a diagnostic one?"** HIV testing of every admitted child of unknown status is policy (screening); TB tests are diagnostic in a symptomatic child.
- **"Am I testing the child or treating my anxiety?"** Defensive "panels" do not protect children; good clinical reassessment does.
- **"Who will follow up this result?"** A test whose result nobody reads is pure harm. Write pending results in the discharge plan.
- **Common student mistakes**: ordering a Widal test for fever; ordering a CXR for every cough; repeating smears daily after an RDT-confirmed malaria; doing an HIV antibody test on a 9-month-old and calling it negative or positive; sending bag urine for culture; doing a sickling test on a 3-month-old; forgetting glucose in a convulsing child; delaying antibiotics until after a CT in meningitis.

# Clinical workup

This is the stepwise approach to deciding which investigations a sick child needs, in the order you would really do it on the ward.

## Step 0: First 5 minutes

1. **Triage (ETAT)**: emergency signs (airway, breathing, circulation/shock, coma, convulsions, severe dehydration) → treat now.
2. **Bedside tests that change emergency treatment**: **$SpO_2$** (oxygen if <90%), **glucose** (dextrose if low), **Hb/HemoCue** (blood if very low), **malaria RDT** (artesunate if severe malaria).
3. **Take samples at the time of cannulation** (one stab): blood culture (if available and it will not delay antibiotics), a small EDTA tube (FBC/cross-match) and a plain tube (chemistry), then start treatment.
4. **Do not send the child for imaging** (CT, X-ray) until stabilised.

## Step 1: Focused history

- [ ] **Duration and pattern of illness**: acute vs chronic tests.
- [ ] **Danger signs**: decide bedside tests now.
- [ ] **Previous treatment and tests** (antimalarials, antibiotics, recent transfusion): affects RDT, cultures, CSF and Hb interpretation.
- [ ] **HIV status** (mother and child), **TB contact**, **immunisation**, **nutrition**: decide HIV PCR/RDT, TB work-up, malnutrition labs.
- [ ] **Family history**: sickle cell disease, bleeding disorders, G6PD deficiency (neonatal jaundice).
- [ ] **What the family can afford** and how far they live (can they return for results?).

## Step 2: Focused examination

- **General**: pallor, jaundice, oedema, wasting (MUAC), lymphadenopathy, hepatosplenomegaly, rash.
- **Focus of infection** (ears, throat, chest, abdomen, urine, joints, skin, meninges).
- **Neurological**: consciousness (AVPU/Blantyre coma score), neck stiffness, fontanelle, focal signs, papilloedema (LP decision).
- **Key negatives to document**: "no neck stiffness, fontanelle flat, no focal signs" justifies not doing an LP; "no chest signs, $SpO_2$ 98%" justifies no CXR.

## Step 3: Bedside tests

| Test | When | Interpretation tips |
|---|---|---|
| **Pulse oximetry** | Every child with cough, difficult breathing, or any danger sign | **<90% → oxygen** (WHO); <94% in emergencies (ETAT); poor trace with cold hands, nail polish, movement |
| **Blood glucose (glucometer)** | Convulsions, coma, lethargy, SAM, neonates, severe malaria, quinine, sepsis | Treat <2.5 (WHO) / <3.0 (UCG) mmol/L; neonates <2.6 |
| **Hb (HemoCue) or PCV** | Pallor, severe malaria, sickle cell crisis, before transfusion | Severe anaemia: Hb <5 (WHO severe malaria), transfusion thresholds Hb <4 or <6 with distress |
| **Malaria RDT** | Every febrile child in Uganda (test before treat) | HRP2 stays positive 2–5 weeks after treatment; *hrp2/3* deletions cause false negatives in some regions |
| **Urine dipstick** | Febrile child ≥3 months without a focus, older children with urinary symptoms, nephrotic/nephritic syndrome, DKA | Nitrite + leucocyte esterase positive → likely UTI (send culture); both negative → UTI unlikely in a verbal child; protein 3+ with oedema → nephrotic; glucose and ketones → DKA |

| Test | When | Interpretation tips |
|---|---|---|
| **HIV rapid test** | All children of unknown status ≥18 months, and mothers of younger children | Positive in a child <18 months shows **exposure**, not infection: send DNA PCR |
| **Bedside clotting test / bleeding time** | Snakebite, suspected DIC | Clot not formed in 20 min → coagulopathy |
| **Transcutaneous bilirubin (TcB)** | Visible neonatal jaundice | Confirm with serum bilirubin if near treatment threshold |
| **MUAC** | Every child 6–59 months | <11.5 cm SAM; 11.5–12.5 MAM |

## Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Malaria microscopy (thick and thin films)** | Severe malaria, suspected treatment failure, recurrent fever within 4 weeks of treatment, species identification (HC III and above) | Parasite density; **>10% parasitaemia or >250,000/µL = hyperparasitaemia** (UCG 2023; WHO severe malaria definition >10%); gametocytes; non-falciparum species | Confirms; monitors clearance (fall by 75% at 48 h); a positive smear after day 3 suggests failure |
| **Full blood count + film** | Severe pallor, bleeding/bruising, suspected leukaemia, sepsis in neonates, HIV on AZT, fever >7 days (HC IV and above) | Microcytic anaemia (iron deficiency), macrocytosis (folate/B12, AZT), blasts (leukaemia), sickle cells, thrombocytopenia (malaria, sepsis, ITP) | Transfusion, iron, referral to haematology/oncology |
| **Blood culture** | Suspected sepsis, meningitis, typhoid, febrile neonates and young infants (regional/national referral; some HC IV) | Organism + sensitivity; CoNS in one bottle is usually contamination | Narrow or change antibiotics; duration |
| **CRP** | Neonatal sepsis (serial), fever without source in infants (where available) | <10 mg/L normal; serial normal CRP at 24–48 h helps stop antibiotics in neonates | Rule-out tool; never use a single CRP to withhold antibiotics from a sick child |
| **Lumbar puncture (CSF)** | Suspected meningitis without contraindication; febrile infants <1 month; prolonged/focal febrile seizures | See CSF table below | Antibiotic choice and duration (bacterial 10–21 days), TB meningitis treatment, cryptococcal antigen in HIV |
| **Urine microscopy and culture** | Febrile infants <3 months; suspected UTI with positive dipstick; recurrent UTI | ≥$10^5$ CFU/mL single organism (clean catch); ≥$10^4$ (catheter); any growth (suprapubic) | Antibiotic choice; imaging for atypical/recurrent UTI |
| **Electrolytes, urea, creatinine** | Severe dehydration, SAM with complications (where available), meningitis, DKA, AKI, severe malaria, prolonged IV fluids | Na <130 (SIADH, hypotonic fluids), K abnormalities, high creatinine | Fluid type and rate, potassium replacement, gentamicin dosing |
| **LFTs** | Jaundice beyond the newborn period, hepatomegaly, TB drugs with symptoms, suspected hepatitis | Conjugated vs unconjugated; transaminases | Hepatitis vs haemolysis vs obstruction; stop hepatotoxic drugs |

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Neonatal bilirubin (total serum)** | Jaundice in the first 24 h, near-threshold TcB, prolonged jaundice (>14 days term) | Plot on hour-specific nomogram (AAP 2022) | Phototherapy, exchange transfusion; conjugated >25 µmol/L (or >20% of total) → biliary atresia work-up |
| **HIV DNA PCR (EID)** | HIV-exposed infants at 6 weeks, 9 months and 6 weeks after weaning; any sick exposed infant <18 months | Positive = infected (confirm with a second sample but **start ART immediately**) | ART, cotrimoxazole |
| **Xpert MTB/RIF Ultra** | Presumptive TB (cough >2 weeks, poor weight gain, contact) on **sputum/induced sputum, gastric aspirate, NPA, stool**, lymph node aspirate, CSF | MTB detected ± rifampicin resistance | Start TB treatment (RR-TB → MDR centre); negative does not exclude TB |
| **Urine LF-LAM** | Children with HIV and TB symptoms, advanced HIV disease, or seriously ill | Positive supports TB | Start TB treatment |
| **Chest X-ray** | Severe pneumonia not improving at 48 h, suspected effusion/empyema, pneumothorax, foreign body, TB, heart failure (HC IV and above) | Consolidation, effusion, cavities, hilar nodes, miliary pattern, cardiomegaly | Drainage, TB treatment, cardiac referral; **not routine** for non-severe pneumonia, bronchiolitis or asthma |
| **Ultrasound** | Abdominal pain/mass, intussusception, effusions, liver abscess, cranial US through the fontanelle in neonates, focused cardiac/lung POCUS | Target sign (intussusception), abscess, hydronephrosis | Surgery, drainage, referral; no radiation |
| **CT / MRI head** | Focal neurology, suspected abscess or tumour, trauma meeting decision-rule criteria (e.g. PECARN), not for simple febrile seizures (national/regional referral) | Mass, abscess, haemorrhage, hydrocephalus | Neurosurgery; CT should never delay antibiotics |
| **Sickle cell confirmation** (Hb electrophoresis/HPLC, isoelectric focusing, validated POC test) | Newborn screening, suspected SCD, positive sickling test | HbSS, HbAS, HbSC | Penicillin V, folic acid, hydroxyurea, clinic enrolment |
| **Blood group and cross-match** | Before transfusion | Compatibility | Safe transfusion |
| **Stool microscopy** | Bloody diarrhoea (amoebae), persistent diarrhoea, suspected giardia or worms causing symptoms | Trophozoites, cysts, ova | Metronidazole/albendazole; **not routine** in acute watery diarrhoea |

## Step 5: Putting it together

**Model summary with an investigation plan:** "Brian, a **9-month-old, 7.8 kg** (WFL Z-score −1), fully immunised, HIV-exposed (mother on TLD, suppressed), still breastfeeding, with **fever for 2 days and one 3-minute generalised convulsion**, now alert and playing, no neck stiffness, fontanelle flat, no focus on examination. **Bedside**: glucose 5.2 mmol/L, **RDT positive**, Hb 10.1 g/dL, $SpO_2$ 98%. **Plan**: treat as **uncomplicated malaria**

**with a febrile convulsion** (oral AL 1 tablet, paracetamol), observe; **no LP** (well, alert, no meningeal signs, immunised), **no CT, no electrolytes**; **urine dipstick** is reasonable (infant with fever); check that his **9-month HIV DNA PCR** has been done (he is due) and record it in the plan."

**Worked example 1: interpreting a CSF**

| Result | Child A (2 years) | Child B (3 years, HIV-negative) | Child C (14 months, weeks of fever) |
|---|---|---|---|
| Appearance | Turbid | Clear | Clear/slightly yellow |
| WBC | **1,800**, 90% neutrophils | **80**, 90% lymphocytes | **220**, 80% lymphocytes |
| Protein | **2.1 g/L** | 0.5 g/L | **2.8 g/L** |
| CSF:blood glucose | **0.2** | 0.7 | **0.25** |
| Gram stain | Gram-positive diplococci | None | None (ZN negative) |
| **Interpretation** | **Bacterial meningitis (pneumococcal)** | **Viral meningitis** (or partially treated bacterial: consider history) | **TB meningitis** until proven otherwise (lymphocytes, very high protein, low glucose, subacute) |
| **Action** | Ceftriaxone 100 mg/kg daily × 10–14 days | Supportive; if antibiotics already given before LP, treat as bacterial | TB treatment (2HRZE/10HR) + prednisolone; Xpert Ultra on CSF; CXR; HIV test |

**Worked example 2: an FBC in a pale toddler** A 2-year-old with Hb 6.8 g/dL, **MCV 58 fL**, MCH low, **RDW high**, platelets 520, normal WBC, no blasts, film: hypochromic microcytes and pencil cells. Mentzer index (MCV ÷ RBC count, e.g. 58 ÷ 3.6 = 16) **>13 suggests iron deficiency** (thalassaemia trait usually <13). History: drinks 1.5 L of cow's milk a day, little meat. **Iron deficiency anaemia** (thrombocytosis is common). **Plan**: oral iron 3 mg/kg/day elemental, deworm, reduce cow's milk, recheck Hb in 4 weeks (expect a rise of ≥1 g/dL); no transfusion (well compensated, no distress).

**Worked example 3: a "hyperkalaemia" that wasn't** A well 5-month-old with gastroenteritis has a $K^+$ of **7.4 mmol/L** on a capillary heel-prick sample; ECG normal, child alert. **Interpretation**: **haemolysed capillary sample** (squeezing) until proven otherwise. **Action**: repeat a free-flowing venous sample before any treatment; result 4.6 mmol/L.

# Differential diagnosis

Interpreting common abnormal results: what the result could mean, and the test that separates the possibilities.

| Abnormal result | Main possibilities in a Ugandan child | Distinguishing test / feature |
|---|---|---|
| **Positive malaria RDT** | Acute malaria; recent treated malaria (HRP2 persistence); asymptomatic parasitaemia with another illness | Microscopy (density, ring stages), timing of last treatment, look for another focus |
| **Negative malaria RDT in a febrile child** | Pneumonia, otitis, UTI, meningitis, typhoid, viral illness, measles, TB; rarely *hrp2*-deleted or non-falciparum malaria | Examination for focus, urine, LP if indicated, microscopy if severe |
| **Microcytic anaemia** | Iron deficiency (diet, hookworm), thalassaemia trait, anaemia of chronic disease (TB, HIV), lead poisoning | Ferritin (raised by inflammation), RDW, Mentzer index, Hb electrophoresis |

| Abnormal result | Main possibilities in a Ugandan child | Distinguishing test / feature |
|---|---|---|
| **Normocytic anaemia with high reticulocytes** | Haemolysis (malaria, SCD crisis, G6PD deficiency, post-artesunate), acute bleeding | Film (sickle cells, bite cells), bilirubin, LDH, G6PD (after recovery), DAT |
| **Macrocytic anaemia** | Folate deficiency (SCD, malnutrition), B12 deficiency, AZT, hydroxyurea | Folate/B12 levels, drug history |
| **Thrombocytopenia** | Malaria, sepsis/DIC, HIV, ITP, leukaemia, dengue, hypersplenism | Film, other cell lines (pancytopenia → marrow), clinical context |
| **Very high WBC (>50 × $10^9$/L)** | Leukaemia, pertussis (lymphocytosis), severe bacterial infection | Film (blasts), clinical context; refer if blasts |
| **Hyponatraemia (<130 mmol/L)** | SIADH (meningitis, pneumonia), hypotonic IV fluids, diarrhoea losses, SAM (total body sodium high), adrenal insufficiency | Fluid history, urine sodium/osmolality, volume status |
| **Raised creatinine** | Pre-renal (dehydration), severe malaria AKI, HUS, nephrotoxins (gentamicin, NSAIDs), glomerulonephritis, obstruction | Urine dipstick and microscopy (blood, protein, casts), ultrasound, response to fluids |
| **Proteinuria 3+** | Nephrotic syndrome, glomerulonephritis, fever/exercise (transient), UTI | Urine protein:creatinine ratio, albumin, BP, haematuria |
| **CSF lymphocytosis** | Viral meningitis, TB meningitis, cryptococcal meningitis (HIV), partially treated bacterial meningitis | Protein, glucose ratio, India ink/CrAg, Xpert Ultra on CSF, history of antibiotics |
| **Conjugated hyperbilirubinaemia in an infant** | Biliary atresia, neonatal hepatitis (infection, CMV, syphilis), metabolic disease | Pale stools, dark urine, ultrasound, refer early (Kasai before 60 days) |

# Investigations

The core paediatric tests: what to look for and why.

| Test | What you are looking for | Why (and pitfalls) |
|---|---|---|
| **Malaria RDT (HRP2 / pLDH)** | *P. falciparum* antigen | Fast, cheap, at all levels; HRP2 persists after treatment; pLDH clears faster |
| **Malaria microscopy** | Species, density, stages | Gold standard in the field; depends on the microscopist; needed to monitor severe malaria |
| **HemoCue Hb** | Haemoglobin | Bedside transfusion decisions; capillary samples can read high or low if squeezed |
| **Glucose** | Hypo/hyperglycaemia | Cheapest life-saving test; check every convulsing or unconscious child |
| **FBC and film** | Anaemia type, infection, platelets, blasts | Age-specific normal values; lymphocytosis normal in young children |
| **Blood culture** | Bacteraemia | Take before antibiotics if no delay; 1–3 mL in infants; Salmonella, pneumococcus, *S. aureus*, Gram-negatives common in Uganda |
| **CSF** | Meningitis type | Cell count within 1 h (cells lyse); paired blood glucose |
| **Urine (dipstick, microscopy, culture)** | UTI, glomerular disease, DKA | Collection method matters; bag urine only for dipstick |

| Test | What you are looking for | Why (and pitfalls) |
|---|---|---|
| **Electrolytes/creatinine** | Fluid and renal status | Haemolysis raises K; age-specific creatinine |
| **HIV tests** | Exposure (antibody) vs infection (PCR) | <18 months: PCR; ≥18 months: antibody algorithm |
| **TB tests** | Xpert Ultra (sputum, gastric, NPA, stool), LF-LAM, TST/IGRA, CXR | Paucibacillary disease: negative tests common; use the WHO treatment-decision algorithm |
| **Sickle tests** | Sickling, electrophoresis/HPLC, POC | Sickling test unreliable <6 months and cannot distinguish trait |
| **G6PD** | Deficiency | False normal during acute haemolysis (young cells): retest 2–3 months later |
| **CXR** | Consolidation, effusion, TB, heart size | Inter-observer variation; does not distinguish bacterial from viral pneumonia |
| **Ultrasound** | Abdomen, effusions, cranial (neonates), POCUS | Radiation-free first choice |
| **Widal test** | (Typhoid antibodies) | **Poor sensitivity and specificity in endemic areas: avoid**; use blood culture where available, otherwise treat clinically |

## Normal CSF values and patterns (RCH Melbourne clinical guideline)

| | Neutrophils (× $10^6$/L) | Lymphocytes (× $10^6$/L) | Protein (g/L) | CSF:blood glucose |
|---|---|---|---|---|
| **Normal, <1 month** | 0 (up to 5% of cells may be neutrophils) | <22 | <1.0 | ≥0.6 |
| **Normal, >1 month** | 0 | ≤5 | <0.4 | ≥0.6 |
| **Bacterial meningitis** | 100–10,000 | <100 | >1.0 | <0.4 |
| **Viral meningitis** | <100 | 10–1,000 | 0.4–1.0 | Usually normal |
| **TB meningitis** | <100 | 50–1,000 | 1.0–5.0 | <0.3 |

A **traumatic tap** adds about 1 white cell per 500–1,000 red cells; partially treated bacterial meningitis can look "viral". **Cryptococcal meningitis** (HIV) can have near-normal cells: always do **CrAg** (or India ink) in HIV-positive children with headache or altered consciousness.

## HIV testing in children (Uganda)

| Age / situation | Test | Notes |
|---|---|---|
| HIV-exposed infant, **6 weeks** (or first contact) | **DNA PCR** (DBS via hub) | Birth PCR for high-risk infants where available; POC EID in some sites |
| **9 months** | Second DNA PCR | |
| **6 weeks after breastfeeding stops** | Third DNA PCR (final outcome) | Breastfeeding is advised up to 1 year or beyond with the mother on ART |
| **18 months** | Rapid antibody test | Negative and not breastfed for 6 weeks → HIV-negative |
| Child ≥18 months, unknown status | Antibody algorithm (rapid tests) | Confirm before ART |

| Age / situation | Test | Notes |
|---|---|---|
| Any positive PCR | **Start ART the same day** and send a confirmatory sample | Do not wait for the confirmation |
| Child on ART | **Viral load** at 6 months then 6–12-monthly | VL >1,000 → intensive adherence counselling, repeat after 3 months |

### TB investigations in children (WHO 2022; Uganda NTLP)

- **Xpert MTB/RIF Ultra** is the initial test: **sputum** (expectorated or induced), **gastric aspirate**, **nasopharyngeal aspirate** or **stool** (now widely used in Uganda for young children), plus **lymph node aspirate** or **CSF** for extrapulmonary TB.
- **Urine LF-LAM** for children living with HIV who have TB symptoms, advanced HIV disease or are seriously ill.
- **TST or IGRA** show **infection**, not disease; useful for contacts and supportive evidence.
- **CXR**: hilar or mediastinal nodes, miliary pattern, cavities, effusion.
- Because children have **paucibacillary disease**, many are **bacteriologically negative**: WHO 2022 **treatment-decision algorithms** (symptoms, CXR, risk factors) allow treatment of children <10 years with presumptive pulmonary TB without bacteriological confirmation.

## Management

Management here means **using investigations well in the care of the child**: choosing a test plan, acting on results, and avoiding harm.

### The rational test plan: a stepwise algorithm

1. **Stabilise first** (ETAT). Bedside glucose, Hb, RDT, $SpO_2$ only; treat emergencies.
2. **Form a working diagnosis and two or three differentials**, with rough probabilities.
3. **For each test you are considering, ask**: Is it needed now, later, or never? Will a positive or negative result change management? Is there a cheaper bedside alternative? Can the family afford it? Who will follow up the result?
4. **Take samples in one go**, smallest volumes, correct tubes and labels, with a clear clinical question on the form ("fever 10 days, ?typhoid" is more useful than "FBC please").
5. **Start treatment** that cannot wait.
6. **Review results on the same day**; act on critical results immediately; **repeat implausible results** before acting.
7. **Document** results and their interpretation in the notes; **pending results go on the discharge summary** with who will act on them.
8. **Stop monitoring tests** when the child is improving and the result will no longer change management.

### Rational tests for common presentations

#### Fever without a focus (child 3 months–5 years, alert, no danger signs)

1. **Malaria RDT** (all levels).
2. **Urine dipstick** (and culture if positive or <3 months), especially in girls and infants.
3. **No** routine FBC, CRP, blood culture, CXR or Widal.
4. Fever ≥7 days → FBC, blood culture (typhoid), urine, HIV test, TB screen, examine again (abscess, endocarditis, malignancy).
5. **Infants <3 months** (NICE NG143): FBC, blood culture, CRP, urine microscopy and culture, CXR only if respiratory signs, **LP in infants <1 month** and in older infants who are unwell or have abnormal blood results;

treat with parenteral antibiotics.

**Convulsion with fever**

1. **Glucose** and **malaria RDT** immediately.
2. **LP** if: meningeal signs, **<12 months** with incomplete Hib/PCV immunisation or pretreated with antibiotics (consider), prolonged or focal seizure, not back to normal within 1 hour, or petechial rash (after stabilisation; antibiotics first if delayed).
3. Simple febrile seizure, alert child, immunised: **no CT, no EEG, no routine electrolytes**.
4. Afebrile seizure or status: glucose, sodium, calcium (neonates, rickets), and consider imaging later.

**Cough and fast breathing**

1. **$SpO_2$** (oxygen if <90%), count RR for a full minute.
2. Non-severe pneumonia: **no CXR, no FBC** (WHO 2024). Treat with oral amoxicillin.
3. Severe or not improving at 48 h: **CXR** (effusion, empyema, pneumatocele, TB), **HIV test**, blood culture where available.
4. Cough >2 weeks, weight loss, contact: **TB work-up** (Xpert Ultra on sputum/gastric/NPA/stool, CXR).
5. Bronchiolitis and asthma: **no routine CXR or bloods**.

**Diarrhoea and dehydration**

1. Assess dehydration clinically (Plans A–C).
2. **No routine stool microscopy or culture** in acute watery diarrhoea.
3. **Electrolytes** only in severe dehydration needing IV fluids for long, SAM with complications (where available), altered consciousness, or seizures.
4. **Bloody diarrhoea**: treat empirically for shigella; stool microscopy for amoebae if not improving.
5. Persistent diarrhoea (≥14 days): HIV test, stool microscopy (giardia, cryptosporidium), nutritional assessment.

**Pallor**

1. **HemoCue Hb** now; **malaria RDT**.
2. Severe anaemia with distress → **grouping and cross-match**.
3. FBC and film where available (type of anaemia, platelets, blasts), sickling/electrophoresis if SCD suspected (≥6 months for sickling test), stool for hookworm in older children.
4. Recheck Hb after transfusion and at 2–4 weeks (and after IV artesunate for delayed haemolysis).

**Neonatal jaundice**

1. Visual assessment is unreliable in dark skin: use **TcB** where available and **total serum bilirubin** when near the treatment threshold or jaundiced in the **first 24 h**.
2. Plot by **hours of age** on the nomogram for gestation and risk factors (**AAP 2022**: thresholds are higher than older charts but neurotoxicity risk factors such as prematurity, haemolysis, G6PD deficiency, sepsis, albumin <3.0 g/dL lower them).
3. Jaundice <24 h or rapidly rising: **blood groups (mother and baby), DAT, FBC/film, G6PD**, sepsis screen.
4. Jaundice >14 days (term): **conjugated bilirubin**, stool colour; pale stools → urgent biliary atresia work-up.

**Suspected HIV**

1. Child <18 months: **DNA PCR**; ≥18 months: rapid test algorithm.
2. Positive: start ART, CD4 (advanced disease screening: **CrAg** if CD4 low in adolescents, **TB LAM**), baseline creatinine if TDF.
3. On ART: viral load monitoring per schedule.

### Acting on results

- **Critical results**: act at once, document the time, inform the senior and the family.
- **Unexpected results**: ask whether the result fits the child; check identity, sample quality, timing; **repeat** before major decisions.
- **Normal results in a sick child**: do not be falsely reassured (early sepsis, early meningitis, paucibacillary TB). Reassess clinically.
- **Incidental findings**: record, explain and arrange appropriate follow-up; avoid cascades of further tests unless they change care.
- **Communicating results**: explain to caregivers what the test showed and what it means; HIV and other sensitive results need private counselling and, for adolescents, attention to consent and confidentiality.

### Reducing waste (Choosing Wisely-type rules for paediatrics)

1. **No Widal test** for typhoid in endemic settings.
2. **No routine CXR** for bronchiolitis, asthma or non-severe pneumonia.
3. **No CT or EEG** for a simple febrile seizure; **no CT for minor head injury** in a child at low risk by a decision rule (PECARN).
4. **No daily "routine bloods"** in a stable child; no daily FBC in preterms without indication.
5. **No stool microscopy** for every child with acute watery diarrhoea.
6. **No antimalarials for RDT-negative fever** (test-and-treat), and no repeated smears after a confirmed diagnosis unless severe or failing.
7. **No bag urine for culture**.

## Complications

Complications of investigations (and of not investigating):

- **Iatrogenic anaemia** in neonates and small children from repeated sampling.
- **Pain, fear and loss of trust**; failed venepuncture attempts; haematomas; nerve injury (femoral sampling in infants).
- **LP complications**: post-LP headache (older children), back pain, rarely **coning** if done with raised ICP, infection, bleeding with low platelets.
- **Contrast reactions and sedation risks** for imaging (aspiration, apnoea).
- **Radiation-induced cancer risk** with repeated CT.
- **False-positive cascades**: unnecessary antibiotics (Widal, contaminated cultures), antimalarials (persistent HRP2), anxiety.
- **False-negative reassurance**: missed meningitis, TB or UTI because a single test was "normal".
- **Financial harm** to families (catastrophic out-of-pocket costs), delayed care-seeking next time.
- **Lost results**: EID results that never reach the family → infected infants not started on ART.

## Ward-round & exam pearls

**Model case presentation (with investigation reasoning):** "Aisha, a **3-year-old girl, 12.5 kg** (WFH Z-score 0, height-for-age –1.5), fully immunised, HIV-unexposed, with **fever for 9 days**, abdominal pain and loose stools, no cough. She looks unwell, temperature 39.2°C, HR 140, mild hepatosplenomegaly, no rash, no neck stiffness. **RDT negative twice**, Hb 10.2 g/dL, urine dipstick normal. **Working diagnosis: typhoid fever** (differentials: UTI

excluded by dipstick, TB, HIV, brucellosis, malignancy). **Tests**: **blood culture** (the most useful test for typhoid; Widal not requested), FBC with film, HIV rapid test; TB symptom screen negative. Started on ceftriaxone 80 mg/kg daily pending culture."

**Viva questions with answers**

1. *"Define sensitivity and specificity, and explain SnOut and SpIn."* Sensitivity = true positives ÷ all with disease; specificity = true negatives ÷ all without disease. A highly sensitive test that is negative rules out; a highly specific test that is positive rules in.
2. *"Why does the PPV of a malaria RDT fall in Kampala?"* PPV depends on prevalence; with low prevalence, false positives make up a larger share of positive results.
3. *"Why can't you use an HIV antibody test in a 9-month-old?"* Maternal IgG antibodies cross the placenta and persist up to 18 months; a positive antibody test shows exposure, not infection. Use DNA PCR.
4. *"What are the contraindications to LP?"* Signs of raised ICP (papilloedema, focal signs, abnormal posture, unequal pupils, irregular breathing), deep or deteriorating coma, shock, coagulopathy/thrombocytopenia, local skin infection. Give antibiotics first.
5. *"Why is the Widal test unhelpful in Uganda?"* Background antibody levels from previous exposure and cross-reacting infections give false positives; early disease gives false negatives; single titres are uninterpretable.
6. *"Which test for TB in a 2-year-old who cannot produce sputum?"* Xpert Ultra on a gastric aspirate, nasopharyngeal aspirate or **stool**; plus CXR, TST/IGRA, HIV test; treat by algorithm if negative but clinically likely.
7. *"How much blood can you take from a preterm baby?"* Total blood volume is about 80–90 mL/kg; keep sampling to a minimum (about ≤1–5% of blood volume per 24 h), batch tests and use microsamples.
8. *"What does a $K^+$ of 7.5 in a well child mean?"* Probably a haemolysed or squeezed sample: repeat a free-flowing sample and check an ECG before treating.

**OSCE tips**

- In a "request the investigations" station, **group** your tests: bedside now, laboratory today, imaging if indicated, and say **why** each changes management.
- Always mention **glucose** in any convulsing, unconscious or severely ill child.
- Interpret results **with the age** of the child ("Hb 9.8 at 8 weeks is physiological").
- In a CSF interpretation station, comment on **appearance, cells (type), protein, glucose ratio, Gram/ZN/India ink**, then give the diagnosis and treatment.

# Mnemonics & memory aids

- **"SnOut, SpIn"**: Sensitive test Negative rules Out; Specific test Positive rules In.
- **"Bedside four in every sick child: G-H-O-M"**: **G**lucose, **H**b, **O**xygen saturation, **M**alaria RDT.
- **"Will it change what I do?"** (the one question before every test).
- **HIV EID "6-9-6-18"**: PCR at **6** weeks, **9** months, **6** weeks after breastfeeding stops; rapid test at **18** months.
- **CSF "5-0-0.4-0.6"** after 1 month: ≤**5** lymphocytes, **0** neutrophils, protein <**0.4** g/L, glucose ratio ≥**0.6**.
- **LP contraindications "COAGS"**: **C**oma (deep/deteriorating), **O**edema of the disc/raised ICP signs, **A**bnormal clotting or platelets, **G**one into shock, **S**kin infection at the site.
- **Sample errors "HEAD"**: **H**aemolysis (high K), **E**DTA contamination (low Ca, high K), **A**rm with the drip (dilution, high glucose), **D**elay (low glucose).

# High-yield summary

## Numbers & doses to know

| Item | Value |
|---|---|
| Hypoglycaemia thresholds | <2.5 mmol/L (WHO), <3.0 (UCG, SAM); neonate <2.6 |
| Oxygen threshold | $SpO_2$ **<90%** → oxygen (WHO); ETAT emergencies <94% |
| Anaemia (WHO 2024 Hb cut-offs) | <10.5 g/dL (6–23 months); <11.0 (24–59 months); <11.5 (5–11 y); <12.0 (12–14 y) |
| Physiological nadir | Hb **9–11 g/dL at 2–3 months** |
| Transfusion threshold (UCG/WHO) | Hb **<4 g/dL**, or **4–6 with distress/complications** |
| Hyperparasitaemia (UCG 2023 / WHO) | **>10%** of RBCs or >250,000/µL |
| HRP2 RDT persistence | **2–5 weeks** after treatment |
| Normal CSF >1 month | ≤5 lymphocytes, 0 neutrophils, protein <0.4 g/L, glucose ratio ≥0.6 |
| Normal CSF <1 month | <22 cells, protein <1.0 g/L |
| Bacterial CSF | Neutrophils 100–10,000, protein >1.0, ratio <0.4 |
| UTI culture thresholds | $\geq 10^5$ CFU/mL (clean catch); $\geq 10^4$ (catheter); any growth (suprapubic) |
| Blood culture volume | **1–3 mL** in infants (≥1 mL); 4–6 mL older children |
| Blood volume | Neonate **80–90 mL/kg**; child 80 mL/kg |
| EID schedule (Uganda) | PCR 6 weeks, 9 months, 6 weeks post-weaning; rapid test 18 months |
| Viral load failure threshold | **>1,000 copies/mL** (twice, 3 months apart with adherence support) |
| Glucose fall in an unseparated sample | ≈5–7% per hour |
| Head CT radiation | ≈1–2 mSv (6–12 months of background) |

## Classic exam traps

- **Trap:** "Negative malaria RDT, but give AL anyway because it is Uganda." **Right:** trust a negative RDT in a non-severe child and look for another cause.
- **Trap:** "Repeat the RDT 1 week after treatment to confirm cure." **Right:** HRP2 persists for weeks; use microscopy if you need to check clearance.
- **Trap:** "HIV rapid test negative at 4 months, so the breastfed infant is HIV-negative." **Right:** use DNA PCR <18 months, and the final status is only known 6 weeks after breastfeeding stops.
- **Trap:** "Order a Widal test for prolonged fever." **Right:** blood culture where available; otherwise clinical diagnosis.
- **Trap:** "CT head before LP in every child with suspected meningitis." **Right:** CT only if clinical signs of raised ICP or focal signs; never delay antibiotics.
- **Trap:** "Bag urine culture grew mixed organisms: treat UTI." **Right:** bag samples are contaminated; collect clean-catch, catheter or suprapubic urine.
- **Trap:** "Sickling test negative at 3 months, so no sickle cell disease." **Right:** HbF hides HbS in early infancy; use electrophoresis/HPLC or a validated POC test.
- **Trap:** "CXR for every child with fast breathing." **Right:** not for non-severe pneumonia, bronchiolitis or asthma; reserve for severe, non-resolving or complicated cases and TB.

## Questions seniors ask

1. **"Why did you order this test?"** State the question it answers and how each result would change your plan.

2. **"Which bedside tests does every sick child need?"** Glucose, Hb, malaria RDT, $SpO_2$ (and HIV test if status unknown).
3. **"How do you know this RDT-positive child has malaria and not something else?"** You don't always: parasitaemia can coexist with pneumonia, meningitis or sepsis; examine fully and treat both if needed.
4. **"When would you do an LP in a child with a febrile convulsion?"** Meningeal signs, prolonged/focal seizure, not back to normal, under 12 months with incomplete immunisation or prior antibiotics, or a petechial rash, once stable and without contraindications.
5. **"How do you diagnose TB in a 2-year-old?"** Contact history, symptoms, growth failure, CXR, Xpert Ultra on gastric aspirate/NPA/stool, TST/IGRA, HIV test; treat by the WHO algorithm if tests are negative but suspicion is high.
6. **"This potassium is 7.2. What will you do?"** Check the child and the ECG; if well and the sample was haemolysed, repeat before treating; if ECG changes, treat as hyperkalaemia now.
7. **"What results are pending at discharge, and who will follow them up?"** Name each (cultures, EID PCR, Xpert, histology) and the person responsible.
8. **"How much would these tests cost the family, and are they worth it?"** Know the main costs locally and prioritise tests that change treatment.

SELF-TEST

# Flashcards

35 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | The one question to ask before ordering any test | **Will the result change my management?** If neither a positive nor a negative result would change what you do, do not order it. |
| 2 | Define sensitivity | True positives ÷ all who **have** the disease. A highly sensitive test, when **negative**, rules the disease out (**SnOut**). |
| 3 | Define specificity | True negatives ÷ all who **do not have** the disease. A highly specific test, when **positive**, rules the disease in (**SpIn**). |
| 4 | What happens to PPV when prevalence falls? | **PPV falls** (more of the positives are false); **NPV rises**. Same test, different meaning in Kampala vs a high-transmission district. |
| 5 | Likelihood ratio formulas | **LR+ = sensitivity ÷ (1 – specificity)**; **LR– = (1 – sensitivity) ÷ specificity**. LR+ >10 or LR– <0.1 changes probability a lot. |
| 6 | Bedside tests for every sick child (G-H-O-M) | **Glucose, Hb (HemoCue), Oxygen saturation, Malaria RDT** (plus HIV test if status unknown). |
| 7 | How long do HRP2 malaria RDTs stay positive after treatment? | About **2–5 weeks** (sometimes longer). Use **microscopy** to judge treatment response or recurrent fever after recent treatment. |
| 8 | Febrile child, negative malaria RDT, no danger signs: what next? | **Do not give antimalarials.** Look for another cause (ears, throat, chest, urine, meninges, skin, measles, typhoid). |
| 9 | Why can't an antibody test diagnose HIV in a 9-month-old? | **Maternal IgG** crosses the placenta and persists **up to 18 months**: a positive antibody test shows exposure. Use **DNA PCR**. |
| 10 | Uganda EID testing schedule | DNA PCR at **6 weeks**, **9 months**, and **6 weeks after breastfeeding stops**; rapid antibody test at **18 months**. Start ART the same day as a positive PCR. |
| 11 | Initial TB test in a child who cannot produce sputum | **Xpert MTB/RIF Ultra** on a **gastric aspirate, nasopharyngeal aspirate or stool** (plus CXR, TST/IGRA, HIV test). Urine **LF-LAM** if HIV-positive. |
| 12 | Does a negative Xpert exclude TB in a child? | **No.** Childhood TB is paucibacillary; use the **WHO 2022 treatment-decision algorithm** (symptoms, CXR, risk) to treat without confirmation. |
| 13 | Normal CSF after 1 month | **≤5 lymphocytes, 0 neutrophils** × $10^6$/L; protein **<0.4 g/L**; CSF:blood glucose **≥0.6**. |
| 14 | Normal CSF in a neonate (<1 month) | Cells **<22** × $10^6$/L (a few neutrophils allowed); protein **<1.0 g/L**; glucose ratio ≥0.6. |
| 15 | CSF pattern of bacterial meningitis | Neutrophils **100–10,000**, protein **>1.0 g/L**, glucose ratio **<0.4**, turbid; Gram stain may show organisms. |
| 16 | CSF pattern of TB meningitis | Lymphocytes 50–1,000, **protein 1–5 g/L**, glucose ratio **<0.3**; subacute illness; Xpert Ultra on CSF; HIV test. |
| 17 | Contraindications to LP ('COAGS') | **C**oma (deep/deteriorating), **O**edema of disc / raised ICP signs (focal signs, unequal pupils, abnormal posture), **A**bnormal clotting/platelets, **G**one into shock, **S**kin infection at site. Give antibiotics first. |
| 18 | UTI culture thresholds | **≥$10^5$ CFU/mL** single organism (clean catch); **≥$10^4$** (catheter); **any growth** (suprapubic aspirate). Never culture bag urine. |
| 19 | Urine testing in infants under 3 months (NICE NG224) | Send **microscopy and culture**; do not rely on dipstick. Nitrite is often negative because urine does not stay long in the infant bladder. |

| # | Question | Answer |
|---|---|---|
| 20 | WHO 2024 Hb cut-offs for anaemia in children | **<10.5 g/dL (6–23 months); <11.0 (24–59 months); <11.5 (5–11 years); <12.0 (12–14 years)**. Severe: <7.0 under 5 years. |
| 21 | Physiological anaemia of infancy | Hb falls to a **nadir of about 9–11 g/dL at 2–3 months** (term) as HbF-rich cells are replaced; no treatment needed in a well infant. |
| 22 | Lymphocyte predominance in children | Lymphocytes outnumber neutrophils from about **1 week to 4–5 years** of age: do not call a normal toddler's differential 'viral'. |
| 23 | Why is ALP high in children? | **Bone growth** (osteoblasts); highest in infancy and the pubertal growth spurt (up to 2–4× adult). Very high with low phosphate → rickets. |
| 24 | Total blood volume in a neonate | **80–90 mL/kg**. A 1 kg preterm has ~85 mL of blood; keep samples to the minimum (≈1–5% of blood volume per 24 h). |
| 25 | Common causes of a falsely high potassium | **Haemolysis** (squeezed heel prick, small needle), **EDTA contamination**, delayed separation, high platelets/WBC. |
| 26 | Why measure glucose promptly? | Red cells keep using glucose: it falls **≈5–7% per hour** in an unseparated sample. Use a bedside meter or fluoride tube. |
| 27 | Why avoid the Widal test? | Endemic background antibodies and cross-reactions → **false positives**; early disease → **false negatives**. Use blood culture or treat clinically. |
| 28 | Sickling test limitations | False negative **under 6 months** (HbF), and cannot tell **trait from disease**. Confirm with **Hb electrophoresis/HPLC** or a validated POC test. |
| 29 | When is a CXR NOT indicated? | **Non-severe pneumonia, bronchiolitis, asthma**, simple croup. Reserve for severe/non-resolving pneumonia, suspected effusion, TB, foreign body or heart failure. |
| 30 | Investigations for a simple febrile seizure | **Glucose and malaria test**, look for the focus; **no CT, EEG or routine electrolytes**. LP if meningeal signs, focal/prolonged, not recovering, or <12 months with incomplete Hib/PCV or prior antibiotics. |
| 31 | Febrile infant under 3 months (NICE NG143) | **FBC, blood culture, CRP, urine microscopy and culture**, CXR only if respiratory signs, stool culture if diarrhoea; **LP if <1 month** or unwell/abnormal bloods; parenteral antibiotics. |
| 32 | G6PD testing pitfall | During or soon after **acute haemolysis** the result can be falsely normal (young red cells have more enzyme): **repeat after 2–3 months**. |
| 33 | A $K^+$ of 7.6 mmol/L in a well, playful child: what do you do? | Suspect **haemolysis or a bad sample**; check ECG, **repeat a free-flowing venous sample** before treating. |
| 34 | HIV viral load: failure threshold | **>1,000 copies/mL** on two tests 3 months apart, with **intensive adherence counselling** after the first. |
| 35 | Head CT radiation dose | About **1–2 mSv**, ≈6–12 months of background radiation; children are more radiosensitive. Use clinical decision rules (PECARN) and ultrasound where possible. |

SELF-TEST

# Single best answer questions

17 questions. The answers and explanations follow at the end.

**1.** A 2-year-old boy (12 kg) in Kampala has had fever for 1 day with a runny nose and red throat. He is alert, playing, RR 30/min, $SpO_2$ 98%, no danger signs. A good-quality malaria RDT is negative. What is the MOST appropriate next step?

A Treat symptomatically with paracetamol, counsel on danger signs and review if fever persists beyond 2–3 days

B Request a thick blood film to confirm the negative RDT

C Give artemether–lumefantrine because malaria is common in Uganda

D Request FBC, CRP and blood culture

E Give amoxicillin for 5 days

**2.** A 4-year-old girl completed a 3-day course of AL for RDT-confirmed malaria 10 days ago. She now has fever again with cough and RR 46/min. A repeat HRP2 RDT is positive. Which interpretation is MOST appropriate?

A She has treatment failure and needs quinine

B The RDT may remain positive from the previous infection; do microscopy and look for another cause such as pneumonia

C The positive RDT proves malaria is the cause of her cough

D She has a new P. vivax infection

E The RDT is faulty and should be discarded

**3.** A 9-month-old HIV-exposed infant attends the clinic. His mother is on ART with a suppressed viral load and he is breastfeeding. His 6-week DNA PCR was negative. A rapid HIV antibody test done today by a student is positive. What is the MOST appropriate interpretation and action?

A He is HIV-infected; start ART today

B The antibody test is falsely positive; no further tests are needed until 18 months

C Repeat the antibody test in 1 week

D Stop breastfeeding immediately

E The positive antibody test reflects maternal antibodies; send his scheduled 9-month DNA PCR

**4.** A 2-year-old girl (10 kg) has had cough for 5 weeks, fever, and has lost weight (weight-for-height Z-score −2.4). Her grandmother, who lives with her, is on TB treatment. She cannot produce sputum. Which investigation plan is MOST appropriate at a general hospital?

A Sputum smear microscopy only

B Tuberculin skin test only; treat if positive

C Wait for TB culture results (6–8 weeks) before deciding

D Xpert MTB/RIF Ultra on a gastric aspirate or stool sample, chest X-ray and HIV test

E Widal test and blood culture

**5.** A 3-year-old boy is drowsy with fever and neck stiffness. HR 150/min, capillary refill 2 s, BP normal, pupils equal and reactive, no focal signs, no papilloedema, platelet count unknown. The CT scanner is 2 days' wait. What is the MOST appropriate sequence? Advanced

- A Wait for the CT scan before LP and antibiotics
- B Give ceftriaxone, then do an LP within the next hour if there are no contraindications
- C Do an LP and wait for the CSF result before giving antibiotics
- D Give paracetamol and observe for 24 hours
- E Treat for malaria only after a positive RDT

**6.** CSF from a 14-month-old with 3 weeks of fever, vomiting and a new left sixth nerve palsy shows: clear fluid, WBC 240 × $10^6$/L (85% lymphocytes), protein 2.6 g/L, CSF:blood glucose ratio 0.2, Gram stain negative. What is the MOST likely diagnosis?

- A Tuberculous meningitis
- B Normal CSF for age
- C Viral meningitis
- D Partially treated pneumococcal meningitis
- E Cerebral malaria

**7.** A 7-week-old girl has fever of 38.6°C for 1 day, is feeding a little less, and has no focus on examination. She is alert, HR 160/min, RR 48/min, $SpO_2$ 98%. Malaria RDT negative. Which investigation is MOST important not to omit?

- A Chest X-ray
- B Stool microscopy
- C Abdominal ultrasound
- D Widal test
- E Urine microscopy and culture from a clean-catch or catheter sample

**8.** A 5-month-old boy with gastroenteritis is alert and drinking. A capillary heel-prick sample (difficult, squeezed) shows $K^+$ 7.4 mmol/L, Na 137 mmol/L. His ECG is normal. What is the MOST appropriate next step?

- A Give calcium gluconate 0.5 mL/kg IV immediately
- B Give insulin and dextrose
- C Start salbutamol nebulisers every 20 minutes
- D Repeat a free-flowing venous sample before any treatment
- E Stop all oral fluids

**9.** A 3-month-old girl of a mother with sickle cell trait has a sodium metabisulphite sickling test reported as negative. The father also has the trait. What is the BEST interpretation?

- A She does not have sickle cell disease
- B She has sickle cell trait
- C The sickling test is unreliable at this age because of high HbF; confirm with Hb electrophoresis/HPLC or a validated point-of-care test
- D Repeat the sickling test at 6 weeks
- E Start penicillin V and hydroxyurea now

**10.** A 2-month-old term baby is brought for immunisation. He is feeding well, thriving (weight-for-age Z-score 0) and active. A routine FBC done at a private clinic shows Hb 9.6 g/dL with normal indices for age. The mother is worried. What is the MOST appropriate response?

A Transfuse 20 mL/kg whole blood
B Start oral iron 6 mg/kg/day
C Explain that this is physiological anaemia of infancy, continue breastfeeding and routine care
D Refer to a haematologist for bone marrow examination
E Check the haemoglobin every week

**11.** A 6-year-old girl (19 kg) has fever for 10 days, headache, abdominal pain and a relative bradycardia. RDT and malaria smear are negative; urine dipstick is normal. The laboratory offers a Widal test or blood culture (5-day result). Which is the MOST rational approach?

A Widal test and treat only if titres are raised
B Blood culture, then start empirical treatment for typhoid (e.g. ceftriaxone or azithromycin) while awaiting the result
C No tests; give antimalarials
D Stool microscopy for ova and cysts
E Wait for the blood culture result before treating

**12.** A 4-year-old (16 kg) with cough and fever for 2 days has RR 44/min, mild chest indrawing, no danger signs and $SpO_2$ 96%. The intern requests a CXR, FBC and CRP 'to confirm pneumonia'. What is the BEST advice?

A Agree: all pneumonia needs radiological confirmation
B Do only the CXR because it distinguishes viral from bacterial pneumonia
C These tests will not change management: treat with oral amoxicillin 40 mg/kg twice daily for 5 days and review
D Do a blood culture before any antibiotics
E Request sputum Xpert for TB

**13.** In a district where 40% of febrile children have malaria, an RDT with sensitivity 95% and specificity 90% is used. In a district where only 4% have malaria, the same test is used. Which statement is TRUE? Advanced

A The positive predictive value is lower in the low-prevalence district
B The sensitivity of the test is lower in the low-prevalence district
C The negative predictive value is lower in the low-prevalence district
D The specificity is higher in the high-prevalence district
E The test performs identically in both districts in every respect

**14.** A 1.1 kg preterm baby on day 6 in the special care unit has had daily FBC, CRP, electrolytes and blood gases since birth (about 2.5 mL of blood per day). He is stable on feeds and his Hb has fallen from 15.8 to 10.2 g/dL. What is the MOST appropriate change? Advanced

A Transfuse and continue daily bloods
B Stop routine daily bloods; test only when results would change management, using microsamples and batching
C Switch all tests to arterial samples
D Add daily blood cultures
E Continue but take samples from the IV line

**15.** A 22-month-old (11 kg), fully immunised, had a 3-minute generalised seizure with fever of 39.5°C at home. In the emergency room 40 minutes later he is alert, playful and has no neck stiffness or focal signs. Glucose 5.8 mmol/L; RDT positive. Which plan is MOST appropriate?

A CT head and EEG before discharge

B LP, electrolytes, calcium and FBC

C Treat malaria with AL, give antipyretics for comfort, observe, and counsel the family; no further investigations

D Start phenobarbital maintenance for 6 months

E Admit for IV artesunate and blood culture

**16.** A 3-day-old term baby is noticed to be jaundiced to the face and chest at 50 hours of age. She is breastfeeding well. Mother is blood group O Rh-positive. Which is the MOST appropriate investigation?

A No test: jaundice at 50 hours is always physiological

B Liver function tests and hepatitis serology

C Conjugated bilirubin and abdominal ultrasound

D Total serum (or transcutaneous) bilirubin plotted on an hour-specific nomogram for her gestation and risk factors

E G6PD and Coombs test only, without bilirubin

**17.** A 10-year-old boy with sickle cell disease has fever 38.9°C, chest pain and new crackles at the right base, $SpO_2$ 91% in air. Which investigation is MOST likely to change immediate management?

A Hb electrophoresis

B Sickling test

C Widal test

D Abdominal ultrasound

E Chest X-ray and Hb (with blood culture where available)

## Answers & explanations

**1. A** · Treat symptomatically with paracetamol, counsel on danger signs and review if fever persists beyond 2–3 days

**A** is correct: a negative quality-assured RDT in a non-severe febrile child effectively excludes falciparum malaria (high NPV, especially in low-prevalence Kampala); the likely cause is a viral URTI, so symptomatic care and safety-netting are appropriate. **C** is wrong: WHO and UCG require test-before-treat; antimalarials for RDT-negative fever waste drugs and miss the real cause. **B** is wrong: microscopy adds little after a negative RDT in a well child; reserve it for severe illness. **D** is wrong: blood tests would not change management in a well child with a clear viral focus. **E** is wrong: there are no signs of pneumonia or bacterial infection.

**2. B** · The RDT may remain positive from the previous infection; do microscopy and look for another cause such as pneumonia

**B** is correct: **HRP2 antigen persists for 2–5 weeks** after successful treatment, so a positive RDT 10 days later does not prove active malaria; **microscopy** shows whether parasites are present, and her fast breathing suggests **pneumonia**. **A** is wrong: failure should be confirmed by microscopy, and the second-line treatment would be another ACT (e.g. DP), not quinine. **C** is wrong: parasitaemia (or antigenaemia) can coexist with another illness. **D** is wrong: HRP2 tests detect falciparum, not vivax. **E** is wrong: the result is expected, not a faulty test.

**3. E** · The positive antibody test reflects maternal antibodies; send his scheduled 9-month DNA PCR

**E** is correct: maternal IgG persists up to 18 months, so a positive antibody test in an exposed infant shows **exposure**, not infection; the Uganda EID algorithm calls for a **DNA PCR at 9 months**, then 6 weeks after breastfeeding stops, and a rapid test at 18 months. **A** is wrong: ART is started on a positive PCR, not an antibody test, at this age. **B** is wrong: the test is not false; it just answers a different question, and he still needs his 9-month PCR. **D** is wrong: breastfeeding with a suppressed mother on ART is recommended. **C** is wrong: a repeat antibody test will stay positive for months.

**4. D** · Xpert MTB/RIF Ultra on a gastric aspirate or stool sample, chest X-ray and HIV test

**D** is correct: young children rarely produce sputum; **Xpert Ultra on gastric aspirate, nasopharyngeal aspirate or stool** (WHO 2022; widely used in Uganda) plus **CXR** and **HIV testing** is the right package, and with a household contact and growth faltering she can be treated by the WHO treatment-decision algorithm even if Xpert is negative. **A** is wrong: smear is insensitive in paucibacillary childhood TB and she cannot produce sputum. **B** is wrong: TST shows infection, not disease, and alone cannot decide treatment. **C** is wrong: waiting weeks delays treatment in a likely case. **E** is wrong: typhoid tests do not address this picture.

**5. B** · Give ceftriaxone, then do an LP within the next hour if there are no contraindications

**B** is correct: suspected bacterial meningitis is an emergency; **antibiotics must not be delayed** for tests. With no clinical signs of raised ICP, shock or focal neurology, an LP can be done promptly (CSF cells and Gram stain still help even shortly after antibiotics; send blood culture first if available). **A** is wrong: CT is not required before LP without clinical signs of raised ICP and must never delay antibiotics. **C** is wrong: waiting for CSF results before antibiotics increases mortality. **D** is wrong: this child is seriously ill. **E** is wrong: malaria testing is appropriate but does not replace treatment of possible meningitis.

**6. A** · Tuberculous meningitis

**A** is correct: a **subacute** illness with cranial nerve palsy and CSF showing **lymphocytes, very high protein (>1 g/L) and very low glucose ratio (<0.3)** is classic **TB meningitis**. **C** is wrong: viral meningitis has protein <1 g/L and normal glucose. **B** is wrong: normal CSF has ≤5 cells and protein <0.4 g/L after 1 month. **D** is wrong: partially treated bacterial meningitis can be lymphocytic but rarely has a 3-week course with such a low glucose and cranial nerve palsy; consider it if antibiotics were given. **E** is wrong: cerebral malaria has essentially normal CSF.

**7. E** · Urine microscopy and culture from a clean-catch or catheter sample

**E** is correct: **UTI** is the commonest serious bacterial infection in febrile infants without a focus; under 3 months **send microscopy and culture** (NICE NG224/NG143) from a clean-catch or catheter sample, along with FBC, blood culture and CRP where available, and consider LP. **A** is wrong: CXR is indicated only with respiratory signs. **B** is wrong: no diarrhoea. **D** is wrong: Widal is unreliable and irrelevant here. **C** is wrong: ultrasound is for confirmed atypical or recurrent UTI, not first-line.

**8. D** · Repeat a free-flowing venous sample before any treatment

**D** is correct: a squeezed heel-prick sample is commonly **haemolysed**, releasing intracellular potassium; in a well infant with a normal ECG, **repeat a good sample** before treating. **A**, **B** and **C** are wrong: they treat a probably spurious result and carry risks (bradycardia, hypoglycaemia, tachycardia). **E** is wrong: stopping fluids in gastroenteritis risks dehydration and is not indicated.

**9. C** · The sickling test is unreliable at this age because of high HbF; confirm with Hb electrophoresis/HPLC or a validated point-of-care test

**C** is correct: in early infancy **HbF** is still high, so the sickling test can be falsely negative, and it cannot distinguish trait from disease anyway; with two carrier parents (25% risk of HbSS) she needs **electrophoresis/HPLC or a validated POC test**. **A** and **B** are wrong: a negative sickling test at 3 months does not establish either. **D** is wrong: testing earlier is even less reliable. **E** is wrong: treatment starts after confirmation (penicillin V from about 3 months once SCD is confirmed; hydroxyurea per protocol).

**10. C** · Explain that this is physiological anaemia of infancy, continue breastfeeding and routine care

**C** is correct: Hb falls to a **physiological nadir of about 9–11 g/dL at 2–3 months** as fetal red cells are replaced; a thriving infant needs reassurance, not treatment. **A** is wrong: transfusion has risks and is not indicated. **B** is wrong: term breastfed infants have adequate iron stores at this age; treatment doses are not needed. **D** is wrong: there are no abnormal cell lines or symptoms. **E** is wrong: repeated testing takes blood without changing management.

**11. B** · Blood culture, then start empirical treatment for typhoid (e.g. ceftriaxone or azithromycin) while awaiting the result

**B** is correct: **blood culture** is the most useful test for enteric fever (highest yield in the first week, still useful later); a sick child with a typical picture should be **treated empirically** while awaiting results. **A** is wrong: the Widal test has poor sensitivity and specificity in endemic areas and should not guide treatment. **C** is wrong: malaria has been excluded twice. **D** is wrong: stool microscopy does not diagnose typhoid. **E** is wrong: delaying treatment in a 10-day febrile illness risks complications (perforation, encephalopathy).

**12. C** · These tests will not change management: treat with oral amoxicillin 40 mg/kg twice daily for 5 days and review

**C** is correct: non-severe (chest-indrawing without danger signs, $SpO_2$ ≥90%) pneumonia is a **clinical diagnosis** treated with oral amoxicillin (WHO 2024); CXR, FBC and CRP do not change management. **A** is wrong: CXR is reserved for severe, non-resolving or complicated pneumonia. **B** is wrong: CXR cannot reliably distinguish bacterial from viral pneumonia. **D** is wrong: blood cultures are low-yield in non-severe pneumonia. **E** is wrong: a 2-day cough without TB features or contact does not warrant TB testing.

**13. A** · The positive predictive value is lower in the low-prevalence district

**A** is correct: sensitivity and specificity are properties of the test, but **PPV falls as prevalence falls** because false positives make up a larger share of positive results (here roughly 86% PPV at 40% prevalence versus about 28% at 4%). **B** and **D** are wrong: sensitivity and specificity do not change with prevalence (in principle). **C** is wrong: NPV is **higher** in the low-prevalence district. **E** is wrong: predictive values differ.

**14. B** · Stop routine daily bloods; test only when results would change management, using microsamples and batching

**B** is correct: his blood volume is only about **90 mL**; 2.5 mL a day for 6 days is about **15 mL (≈17%)**, a major cause of **iatrogenic anaemia**. Stop routine tests, use capillary microsamples, and batch tests that change management. **A** is wrong: continuing the cause of blood loss is irrational. **C** is wrong: arterial samples do not reduce volume and add risk. **D** is wrong: cultures without clinical suspicion add blood loss and false positives. **E** is wrong: samples from infusing lines are diluted and contaminated.

**15. C** · Treat malaria with AL, give antipyretics for comfort, observe, and counsel the family; no further investigations

**C** is correct: this is a **simple febrile seizure** (6 months–5 years, generalised, <15 min, full recovery, immunised child) triggered by **uncomplicated malaria**; glucose and RDT were the right tests, and no imaging, EEG or LP is needed. **A** is wrong: neuroimaging and EEG are not indicated after a simple febrile seizure. **B** is wrong: LP is not routine in an alert, immunised child over 12 months with no meningeal signs; electrolytes rarely change management. **D** is wrong: prophylactic anticonvulsants are not recommended. **E** is wrong: a single brief seizure with full recovery is not a criterion for severe malaria (two or more convulsions in 24 h are).

**16. D** · Total serum (or transcutaneous) bilirubin plotted on an hour-specific nomogram for her gestation and risk factors

**D** is correct: visible jaundice should be **measured** (TcB, then TSB if near the threshold) and plotted by **hour of age** on the nomogram (AAP 2022), considering neurotoxicity risk factors (prematurity, haemolysis such as ABO incompatibility with an O mother, G6PD deficiency, sepsis). **A** is wrong: visual assessment is unreliable (especially in dark skin) and physiological jaundice can still reach treatment thresholds. **C** and **B** are wrong: conjugated hyperbilirubinaemia work-up is for prolonged jaundice (>14 days) or pale stools. **E** is wrong: DAT/blood group and G6PD may help explain the cause, but the bilirubin level decides treatment.

**17. E** · Chest X-ray and Hb (with blood culture where available)

**E** is correct: suspected **acute chest syndrome** (fever, chest pain, new infiltrate, hypoxia) needs a **CXR** to confirm a new infiltrate and an **Hb** to guide transfusion, plus blood culture; management includes oxygen, antibiotics, analgesia and transfusion. **A** and **B** are wrong: his diagnosis of SCD is already known. **C** is wrong: Widal is not useful. **D** is wrong: abdominal ultrasound does not address his chest.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Prolonged fever in a toddler: choosing tests that matter

A 3-year-old girl (13 kg) is brought to Mengo Hospital with fever for 12 days despite two courses of antimalarials from a drug shop. She has lost weight, has a cough for 3 weeks, and her father was recently diagnosed with smear-positive TB. On examination: temperature 38.6°C, RR 38/min, $SpO_2$ 96%, cervical lymph nodes 1.5 cm, mild hepatosplenomegaly, weight-for-height Z-score −2.1. Immunisation complete; HIV status unknown; mother never tested.

**1. Which bedside tests do you do first, and why?**

> **Malaria RDT** (but she had recent antimalarials, so HRP2 may be positive from earlier infection: if positive, do **microscopy**), **Hb (HemoCue)**, **glucose**, **$SpO_2$** (96%: no oxygen needed), **MUAC**, and an **HIV rapid test** for the mother (and child is ≥18 months, so an antibody test on the child is valid).

**2. RDT positive, microscopy negative for parasites. Interpret.**

> **HRP2 persistence** from the recently treated infection; no current parasitaemia. **Do not give more antimalarials**; look for the real cause.

**3. Which investigations for TB, and which samples?**

> **Xpert MTB/RIF Ultra** on **induced sputum, gastric aspirate or stool** (and node aspirate if a node is accessible), **CXR** (hilar nodes, miliary pattern), **TST/IGRA** if available; HIV test result decides **urine LF-LAM**. Household contact + weight loss + cough >2 weeks = high pre-test probability.

**4. Xpert on stool is negative; CXR shows right hilar lymphadenopathy. What now?**

> A negative Xpert **does not exclude** paucibacillary childhood TB. Using the **WHO 2022 treatment-decision algorithm** (contact, symptoms, CXR), she qualifies for TB treatment: with intrathoracic nodes without airway obstruction she may have **non-severe TB** (4-month 2HRZ(E)/2HR) if criteria are met; otherwise 6 months. Notify, screen other contacts, give nutritional support.

**5. Which tests would be wasteful here?**

> **Widal test** (unreliable), **repeated malaria RDTs**, **routine stool microscopy**, CT scans, broad 'fever panels'. A **blood culture** is reasonable if she becomes sicker (typhoid, invasive non-typhoidal Salmonella), and **FBC with film** if pallor or abnormal counts would change management.

**Take-away:** Recent antimalarials make HRP2 RDTs misleading, and a negative Xpert does not rule out childhood TB: use pre-test probability and the WHO algorithm.

## Case 2: Febrile young infant: a structured work-up

A 5-week-old boy (4.1 kg) is brought to a regional referral hospital with fever 38.7°C for 12 hours and reduced feeding. He is irritable when handled, HR 178/min, RR 58/min, capillary refill 3 s, fontanelle slightly full. Mother is HIV-negative; he received BCG and OPV0. Glucose 4.2 mmol/L, RDT negative.

**1. What are the immediate priorities before tests?**

> ETAT: warmth, **oxygen** if $SpO_2$ <90%, IV access, **fluid bolus only with care** (10–20 mL/kg if shocked), and **do not delay antibiotics**: ampicillin 50 mg/kg + gentamicin 7.5 mg/kg (he is >1 week old) or cefotaxime/ceftriaxone if meningitis suspected and not jaundiced.

**2. Which investigations will you send, and why each?**

> **Blood culture** (bacteraemia; before antibiotics if no delay), **FBC and CRP** (supporting evidence), **urine by catheter or clean catch for microscopy and culture** (UTI common in infants), **LP** (irritable, full fontanelle: meningitis must be excluded), glucose (done). CXR only if respiratory signs.

**3. Is an LP safe now?**

> Check contraindications: **no shock** after resuscitation, no focal signs, no deep coma, platelets/clotting acceptable, no skin infection. If he is unstable, give antibiotics and do the LP when stable; CSF can still be informative.

**4. CSF: WBC 1,200 × $10^6$/L (85% neutrophils), protein 1.8 g/L, glucose ratio 0.25, Gram-negative rods. Interpret and act.**

> **Bacterial meningitis** (Gram-negative, e.g. *E. coli*, *Klebsiella*, Salmonella). Use a **cephalosporin with good CSF penetration** (cefotaxime or ceftriaxone at meningitic doses) ± gentamicin per sensitivity; **treat for at least 21 days** for Gram-negative meningitis; monitor head circumference, seizures, hearing.

**5. Which follow-up tests are rational, and which are not?**

> Rational: culture sensitivities, **sodium** (SIADH) if on IV fluids, **cranial ultrasound** through the fontanelle if not improving (abscess, hydrocephalus), **hearing test** before discharge. Not rational: daily FBC/CRP when he is clearly improving, repeat LP routinely if improving (consider only if not responding or for Gram-negative meningitis per protocol).

> **Take-away:** In febrile young infants, antibiotics come first, then a targeted set of cultures (blood, urine, CSF): each test answers a specific question.

## Case 3: A pale child with a confusing set of results

A 6-year-old boy (18 kg) is referred from an HC IV with 'severe anaemia'. He has had tiredness for 2 months, bone pains and easy bruising. Temperature 37.9°C, HR 128/min, RR 28/min, marked pallor, petechiae on the legs, liver 4 cm and spleen 5 cm below the costal margins, cervical lymph nodes 2 cm. HemoCue Hb 4.6 g/dL. RDT negative.

**1. What is your working diagnosis and immediate plan?**

> Pancytopenic picture (pallor, bruising, petechiae) with organomegaly and bone pain: **acute leukaemia until proven otherwise** (differentials: aplastic anaemia, HIV, visceral leishmaniasis, hypersplenism with chronic malaria). Immediate: group and cross-match, **transfuse** (Hb 4.6 with tachycardia), **platelets** if bleeding.

**2. Which investigations will most change management?**

> **FBC with blood film** (blasts? other cell lines), **HIV test**, **uric acid, potassium, phosphate, calcium, creatinine** (tumour lysis risk), **LDH**, CXR (mediastinal mass in T-cell ALL), and referral for **bone marrow aspirate** at the Uganda Cancer Institute or national referral.

**3. The FBC shows WBC 86 × $10^9$/L with 70% blasts, platelets 18 × $10^9$/L. What precautions in sampling and transfusion?**

> Sample once with minimal volume; **avoid IM injections and arterial punctures** (low platelets); **transfuse slowly** (very high WBC: hyperviscosity; aim for modest Hb rise); hydrate and start **allopurinol** for tumour lysis prevention; urgent oncology referral.

**4. His potassium is reported as 7.1 mmol/L but the ECG is normal. Interpret.**

> With a very high white count (and platelets), **pseudohyperkalaemia** from cell lysis in the tube is common: repeat with a carefully handled sample (heparin, prompt analysis) while watching the ECG. True hyperkalaemia from tumour lysis is also possible: treat if ECG changes or a repeat confirms.

**5. Which tests are unnecessary at this point?**

> Widal, stool microscopy, repeated RDTs, CT scans of multiple regions before diagnosis, and extensive viral serology: none change the urgent plan, which is diagnosis by film and marrow, supportive care and referral.

**Take-away:** When the clinical picture points to a serious diagnosis, choose the tests that confirm it and keep the child safe; interpret odd results (like potassium) in light of the disease.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.